

Database Management Systems

Members:

Aditya Malik (2023046)
Ishaan Raj (2023248)
Rounak Dey (2023449)
Shivam Kumar Jha (2023506)

High-level description of the queries along with result of execution:

1. List all RPG games published by a specific publisher with their ratings and total purchases

	Game_Title	Rating	Publisher_ID	License_Number	Total_Purchases
▶	Mystic Legends	4.3	102	LIC456B	1
	Knight Conquest	4.3	101	LIC123A	0
	Dragon Realm	4.6	102	LIC456B	0
	Heroes of Myth	4.4	102	LIC456B	0

2. Find the publisher with the highest total revenue using subquery

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	Publisher_ID	License_Number	total_revenue
▶	107	LIC901G	69.99

3. Retrieve customers who have spent more than the average order amount

	Customer_ID	Name	Total_Spent
▶	1	Alice Johnson	49.99
	3	Charlie Brown	59.99
	7	Grace Lee	69.99
	10	Jack Turner	54.99
	11	Katherine Brooks	64.99
	12	Liam Johnson	74.99
	15	Olivia Hall	52.99
	16	Paul Wright	63.99
	19	Sophia Baker	59.49

4. List publishers who haven't published any games

	Publisher_ID	License_Number	Type
▶	111	LIC111K	Company
	112	LIC222L	Individual
	113	LIC333M	Company
	114	LIC444N	Individual
	115	LIC555O	Company
	116	LIC666P	Individual
	117	LIC777Q	Company
	118	LIC888R	Individual
	119	LIC999S	Company
	120	LIC000T	Individual
	121	LIC135U	Company
	122	LIC246V	Individual
	123	LIC357W	Company
	124	LIC468X	Individual
	125	LIC579Y	Company
	126	LIC680Z	Individual
	127	LIC791A1	Company
	128	LIC802B1	Individual
	129	LIC913C1	Company
	130	LIC024D1	Individual
	131	LIC135E1	Company
	132	LIC246F1	Individual
	133	LIC357G1	Company
	134	LIC468H1	Individual
	135	LIC579I1	Company
✱	NULL	NULL	NULL

5. Loyal and regular customers

	Customer_ID	Name	Total_Orders	Customer_Type
▶	1	Alice Johnson	1	Regular
	2	Bob Smith	1	Regular
	3	Charlie Brown	1	Regular
	4	David Williams	1	Regular
	5	Emma Davis	1	Regular
	6	Frank White	1	Regular
	7	Grace Lee	1	Regular
	8	Her Henry Adams		Regular
	9	Ivy Clark	1	Regular
	10	Jack Turner	1	Regular
	11	Katherine Br...	1	Regular
	12	Liam Johnson	1	Regular
	13	Mia Roberts	1	Regular
	14	Noah Lewis	1	Regular
	15	Olivia Hall	1	Regular
	16	Paul Wright	1	Regular
	17	Quinn Scott	1	Regular
	18	Ryan King	1	Regular
	19	Sophia Baker	1	Regular
	20	Thomas Green	1	Regular
	21	Uma Nelson	0	Regular
	22	Victor Harris	0	Regular
	23	Wendy Moore	0	Regular
	24	Xavier Perry	0	Regular
	25	Yara Rogers	0	Regular
	26	Zane Bryant	0	Regular
	27	Abigail Fisher	0	Regular
	28	Benjamin Clark	0	Regular
	29	Charlotte Mo...	0	Regular
	30	Daniel White	0	Regular
	31	Evelyn Young	0	Regular
	32	Gabriel Thomas	0	Regular
	33	Hannah Walker	0	Regular
	34	Isaac Allen	0	Regular
	35	Jessica Wilson	0	Regular

6. Identify the top 3 games with the best ratings and most reviews

	Game_ID	Title	Score
▶	10	City Tycoon	5.900000095367432
	5	Brain Teasers	5.800000190734863
	9	Speed Demons	5.699999809265137

7. Find moderators who have resolved below-average issues

	Mod_ID	Genre	Issues_resolved
▶	2	RPG	40
	3	Strategy	30
	4	Adventure	35
	6	Horror	25
•	NULL	NULL	NULL

8. Get all publishers who have never published a game.

	Publisher_id	Type
▶	111	Company
	112	Individual
	113	Company
	114	Individual
	115	Company
	116	Individual
	117	Company
	118	Individual
	119	Company
	120	Individual
	121	Company
	122	Individual
	123	Company
	124	Individual
	125	Company
	126	Individual
	127	Company
	128	Individual
	129	Company
	130	Individual
	131	Company
	132	Individual
	133	Company
	134	Individual
	135	Company

9. List the 5 top-rated blog posts along with their forum title

	Forum_ID	Type	Rating	ratio
►	10	Moderator	4.9	2.3
	5	User	4.8	2.2
	7	Moderator	4.6	2.1
	1	Moderator	4.5	2
	9	Publisher	4.7	2
*	NULL	NULL	NULL	NULL

10. Find the moderators who have resolved fewer than 10 issues but have high engagement (>50).

	Mod_id	Issues_resolved	Engagement
►	2	4	180
	7	5	210
	13	3	155

11. List all customers who have placed an order using UPI as a payment method.

	Customer_id	Name
►	3	Charlie Brown
	7	Grace Lee
	11	Katherine Brooks
	15	Olivia Hall
	18	Ryan King

12. Find all games with a rating above 4 and their corresponding publisher names.

	Title	Rating	Publisher_id
►	Warrior Quest	4.5	101
	Mystic Legends	4.3	102
	Jungle Explorer	4.2	104
	Brain Teasers	4.8	105
	Haunted Mansion	4.1	106
	Battlefield Elite	4.6	107
	Soccer Stars	4.4	108
	Speed Demons	4.7	109
	City Tycoon	4.9	110
	Knight Conquest	4.3	101
	Dragon Realm	4.6	102
	Kingdom Tactics	4.1	103
	Treasure Island	4.7	104
	Logic Masters	4.5	105
	Zombie Nightmare	4.2	106
	Gunfire Glory	4.8	107
	Tennis Titans	4.3	108
	Drift Kings	4.6	109
	Farm Empire	4.7	110
	Alien Invasion	4.5	101
	Heroes of Myth	4.4	102
	Lost in Time	4.3	104
	Number Quest	4.9	105
	Ghost Whisperer	4.1	106
	Warzone Elite	4.7	107

13. Display the total number of orders for each payment method

	Payment_method	Total_Orders
►	Credit	6
	Debit	5
	UPI	5
	PayPal	4

14. Retrieve the names of publishers who have published games in multiple genres.

	Publisher_id
▶	101

15. Find the forum posts that have received more than 5000 impressions.

	Forum_ID	Title	Analytics
▶	16	Monetization Insights	16500
	19	Virtual Reality Gaming	12500

16. List the customers and their total spending amount if they have placed more than 3 orders.

	Customer_id	Name	Total_Spent
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17. Get the average rating of games for each genre.

	Genre	Average_Rating
▶	Action	4.5
	RPG	4.400000095367432
	Strategy	4.0333333015441895
	Adventure	4.399999936421712
	Puzzle	4.733333428700765
	Horror	4.133333206176758
	Shooter	4.699999968210856
	Sports	4.366666793823242
	Racing	4.699999968210856
	Simulation	4.833333333333333

18. Retrieve all games that have not received any reviews.

	Game_ID	Title
►	11	Knight Conquest
	12	Dragon Realm
	13	Kingdom Tactics
	14	Treasure Island
	15	Logic Masters
	16	Zombie Nightmare
	17	Gunfire Glory
	18	Tennis Titans
	19	Drift Kings
	20	Farm Empire
	21	Alien Invasion
	22	Heroes of Myth
	23	Battle Commanders
	24	Lost in Time
	25	Number Quest
	26	Ghost Whisperer
	27	Warzone Elite
	28	Basketball Blitz
	29	Nitro Rush
	30	Space Colony

3) Wrong results by LLM

1) Incorrect Query for Query #1

(Incorrect JOIN condition on Cart table, leading to mismatched data)

```
SELECT
    g.Title AS Game_Title,
    g.Rating,
    p.Publisher_ID,
    p.License_Number,
    COALESCE(SUM(c.Total), 0) AS Lifetime_Purchases
FROM Game g
JOIN Publisher p ON g.Publisher_ID = p.Publisher_ID
LEFT JOIN Cart c ON g.Game_ID = c.Game_ID -- Incorrect: Should be Cart_ID
WHERE g.Genre = 'FPS'
GROUP BY g.Game_ID, g.Title, g.Rating, p.Publisher_ID, p.License_Number;
```

Logical Error:

- **Cart** should be joined using **Cart_ID**, but it mistakenly joins on **Game_ID**.
- This could **lead to incorrect or missing sales data** if **Cart_ID** is the correct field.

Correct query:

```
SELECT
    g.Title AS Game_Title,
    g.Rating,
    p.Publisher_ID,
    p.License_Number,
    COALESCE(SUM(c.Total), 0) AS Lifetime_Purchases
FROM Game g
JOIN Publisher p ON g.Publisher_ID = p.Publisher_ID
LEFT JOIN Cart c ON g.Game_ID = c.Cart_ID
WHERE g.Genre = 'FPS'
GROUP BY g.Game_ID, g.Title, g.Rating, p.Publisher_ID, p.License_Number;
```

2) Incorrect Query for Query #5
(Incorrect classification of customer types)

```
SELECT c.Customer_ID, c.Name, COUNT(o.Order_ID) AS Total_Orders,
       CASE
         WHEN COUNT(o.Order_ID) > 20 THEN 'Loyal' -- Incorrect: Should be >= 20
         ELSE 'Regular'
       END AS Customer_Type
FROM Customer c
LEFT JOIN Orders o ON c.Customer_ID = o.Customer_ID
GROUP BY c.Customer_ID, c.Name;
```

Logical Error:

- Customers with **exactly 20 orders** are incorrectly marked as "Regular" instead of "Loyal".

Correct query:

```
SELECT c.Customer_ID, c.Name, COUNT(o.Order_ID) AS Total_Orders,
       CASE
         WHEN COUNT(o.Order_ID) >= 20 THEN 'Loyal'
         ELSE 'Regular'
       END AS Customer_Type
FROM Customer c
LEFT JOIN Orders o ON c.Customer_ID = o.Customer_ID
GROUP BY c.Customer_ID, c.Name;
```

3. Incorrect Query for Query #6 (Incorrect calculation of game scores)

```
SELECT g.Game_ID, g.Title, g.Rating + COUNT(r.Customer_ID) AS Score
FROM Game g
LEFT JOIN Reviews r ON g.Game_ID = r.Game_ID
GROUP BY g.Game_ID
ORDER BY Score DESC
LIMIT 3;
```

Logical Error:

- **Rating** is typically a **floating-point number**, while **COUNT(r.Customer_ID)** is an **integer**.
- Adding these together **does not make sense mathematically**.

Correct query:

```
SELECT g.Game_ID, g.Title, g.Rating, COUNT(r.Customer_ID) AS Review_Count
FROM Game g
LEFT JOIN Reviews r ON g.Game_ID = r.Game_ID
GROUP BY g.Game_ID
ORDER BY Review_Count DESC, g.Rating DESC
LIMIT 3;
```

4) Incorrect Query for Query #10
(Incorrect self-join condition on Orders table)

```
SELECT DISTINCT o.Customer_id
FROM Orders o
JOIN Game g ON o.customer_id = o.customer_id -- Incorrect: Self-join on same column
LEFT JOIN Reviews r ON g.game_id = r.game_id
AND o.customer_id = r.customer_id
GROUP BY o.customer_id
HAVING COUNT(DISTINCT g.game_id) = COUNT(r.game_id);
```

Logical Error:

- `o.customer_id = o.customer_id` is always true, leading to an incorrect join.
- The self-join should instead be `o.Game_ID = g.Game_ID`.

Correct Query:

```
SELECT distinct o.Customer_id
FROM Orders o join Game g
on o.customer_id = o.customer_id
JOIN join Reviews r on g.game_id = r.game_id and o.customer_id = r.customer_id
GROUP BY o.customer_id
HAVING COUNT(DISTINCT g.game_id) = COUNT(r.game_id);
```

5. Incorrect Query for Query #16

(Using SUM() instead of COUNT() in HAVING clause)

Incorrect Query:

```
SELECT C.Customer_id, C.Name, SUM(O.Total) AS Total_Spent
FROM Customer C
INNER JOIN Orders O ON C.Customer_id = O.Customer_id
GROUP BY C.Customer_id, C.Name
HAVING SUM(O.Total) > 3; -- Incorrect: Should check COUNT of orders, not SUM of
total spent.
```

Logical Error:

- The condition `SUM(O.Total) > 3` checks **spending amount** instead of **number of orders**.
- It should check `COUNT(O.Order_ID) > 3`.

Correct Query:

```
SELECT C.Customer_id, C.Name, SUM(O.Total) AS Total_Spent
FROM Customer C
INNER JOIN Orders O ON C.Customer_id = O.Customer_id
GROUP BY C.Customer_id, C.Name
HAVING COUNT(O.Order_id) > 3;
```